

Constraint Optimization of MicroPlate Designs: supplementary materials

Ramiz Gindullin¹, María Andreína Francisco Rodríguez¹

¹Uppsala University, Sweden
Ramiz.Gindullin@it.uu.se, Maria.Andreina.Francisco@it.uu.se

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Abstract

We present detailed results of computational experiments of the paper published in the proceedings of the 2026 AAAI conference [3].

Overview

Here we provide the complete list of figures and tables containing replicated results of replicated experiments [2] of the paper [1] (we will use ‘the PLAID paper’ for brevity):

- Figures 1 and 2 correspond to Figures 2 and 3 (resp.) of the PLAID paper;
- Figures 3–26 correspond to Figures 1–24 (resp.) from the supplementary materials of the PLAID paper,
- Tables 1–5 correspond to Tables 1–5 (resp.) from the supplementary materials of the PLAID paper.

Figures 1 and Figures 3–17 showcase various results of dose-response experiments.

Figures 2 and Figures 18–26 showcase various results of screening experiments.

References

- [1] María Andreína Francisco Rodríguez, Jordi Carreras Puigvert, and Ola Spjuth. Designing microplate layouts using artificial intelligence. *Artificial Intelligence in the Life Sciences*, 3:100073, 2023.
- [2] Ramiz Gindullin and María Andreína Francisco Rodríguez. COMPD AAAI 2026 evaluation, 2025. Accessed: 2025-10-01.
- [3] Ramiz Gindullin and María Andreína Francisco Rodríguez. Constraint optimization of microplate designs. In *Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence*, 2026. (accepted).

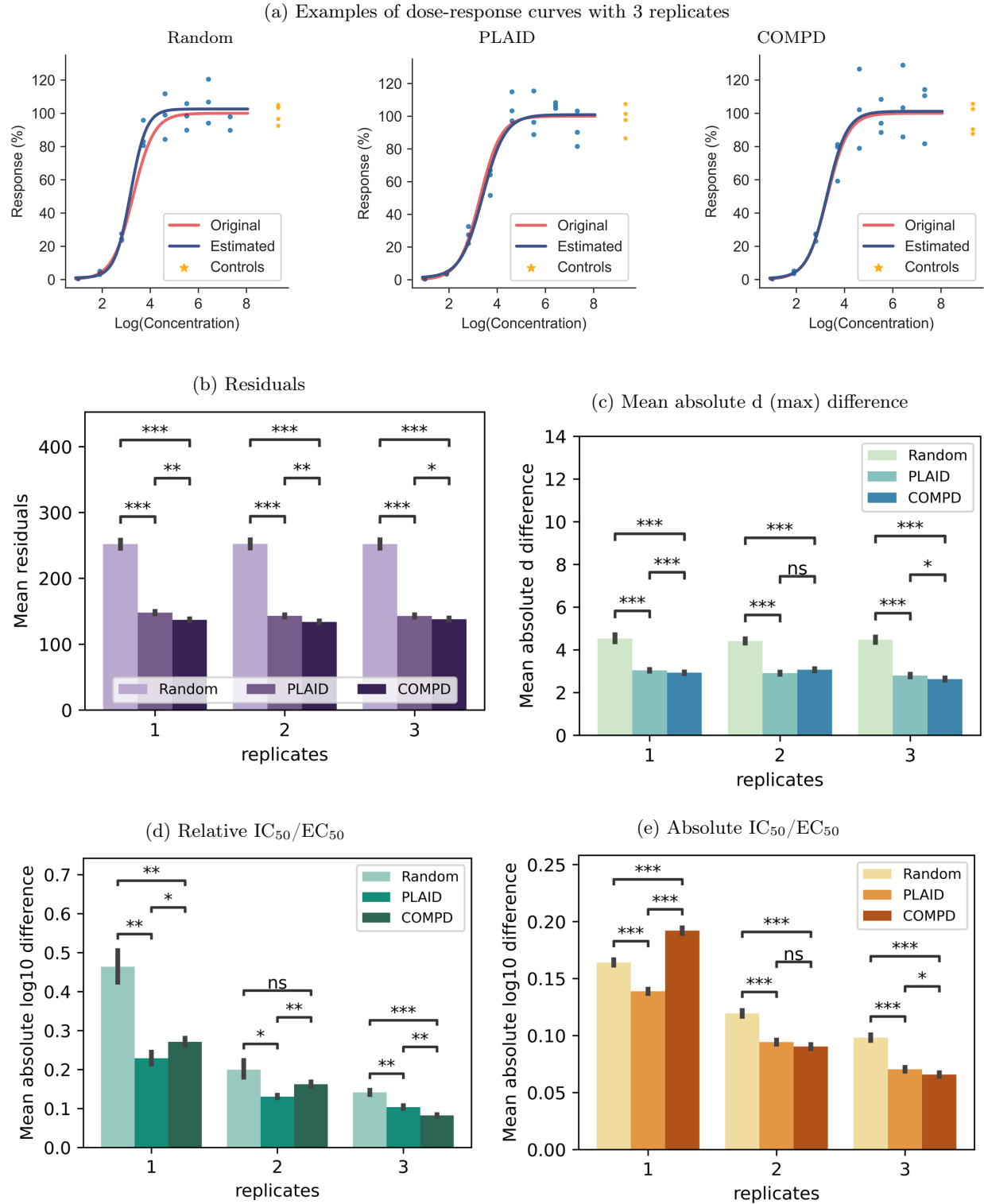


Figure 1: Comparison between expected and obtained values for dose response curves with 8 doses, 1, 2, or 3 replicates, 20 negative controls on 384-well plate, and strong plate effects with a linear relationship to column number on the right side of the plate. * indicates $p < 10^{-4}$, ** indicates $p < 10^{-12}$, *** indicates $p < 10^{-43}$.

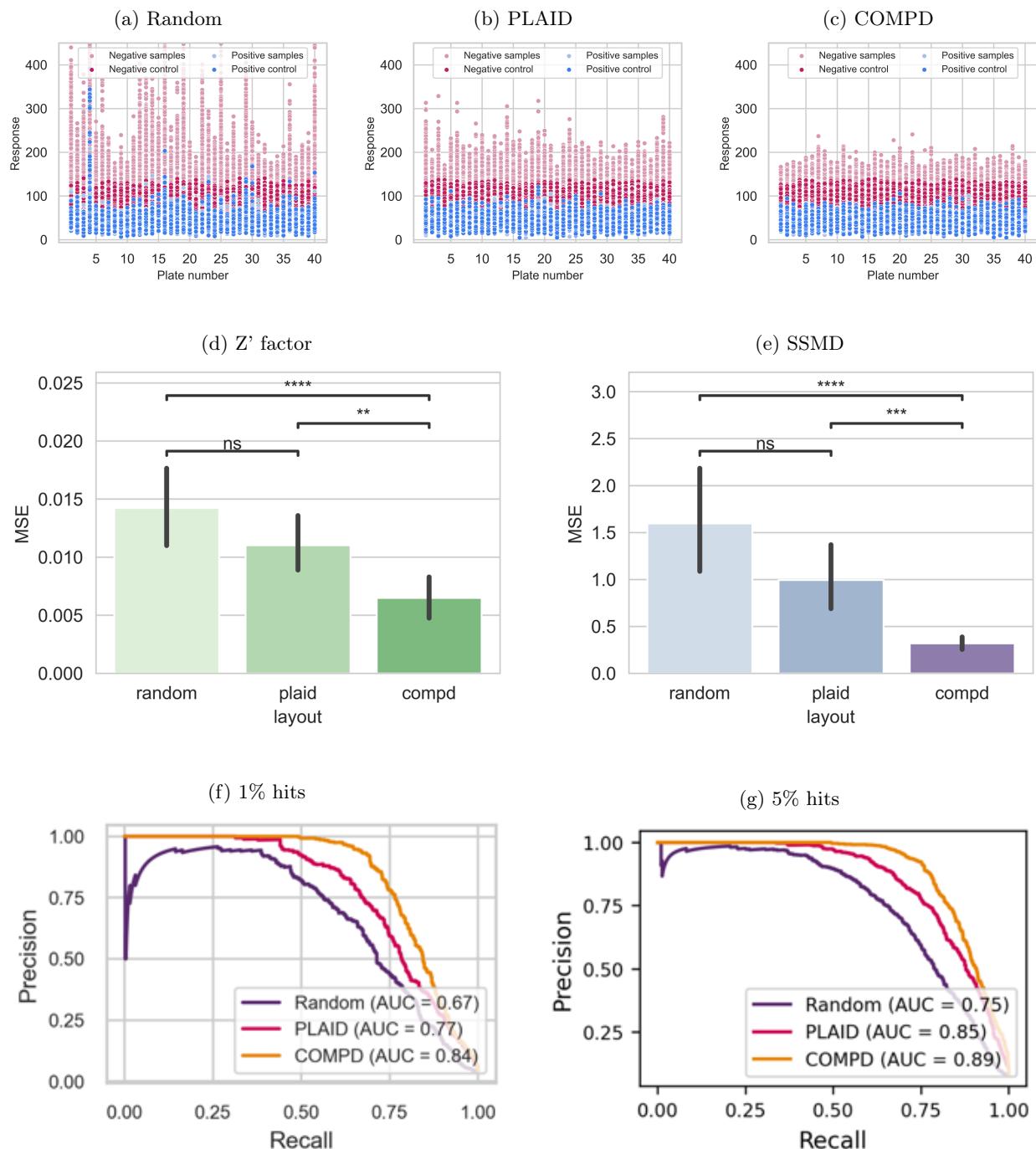


Figure 2: Screening experiments consisting of 40 microplates for each type of layout, each with 10 positive and 10 negative controls and only 1 replicate per compound. Hits are randomly distributed on each plate. (a)-(c): Simulated response data with 1% hit rate after error correction and normalisation in the presence of mild bowl-shaped plate effects. MSE between expected and obtained (d) Z' factor and (e) SSMD in the presence of mild bowl-shaped plate effects. The expected values were calculated using plates randomly filled with 50% positive controls and 50% negative controls, constituting the optimal values obtainable by these metrics. (f) and (g): PR (precision recall) curves for experiments with varying hit rates in the presence of strong bowl-shaped plate effects.

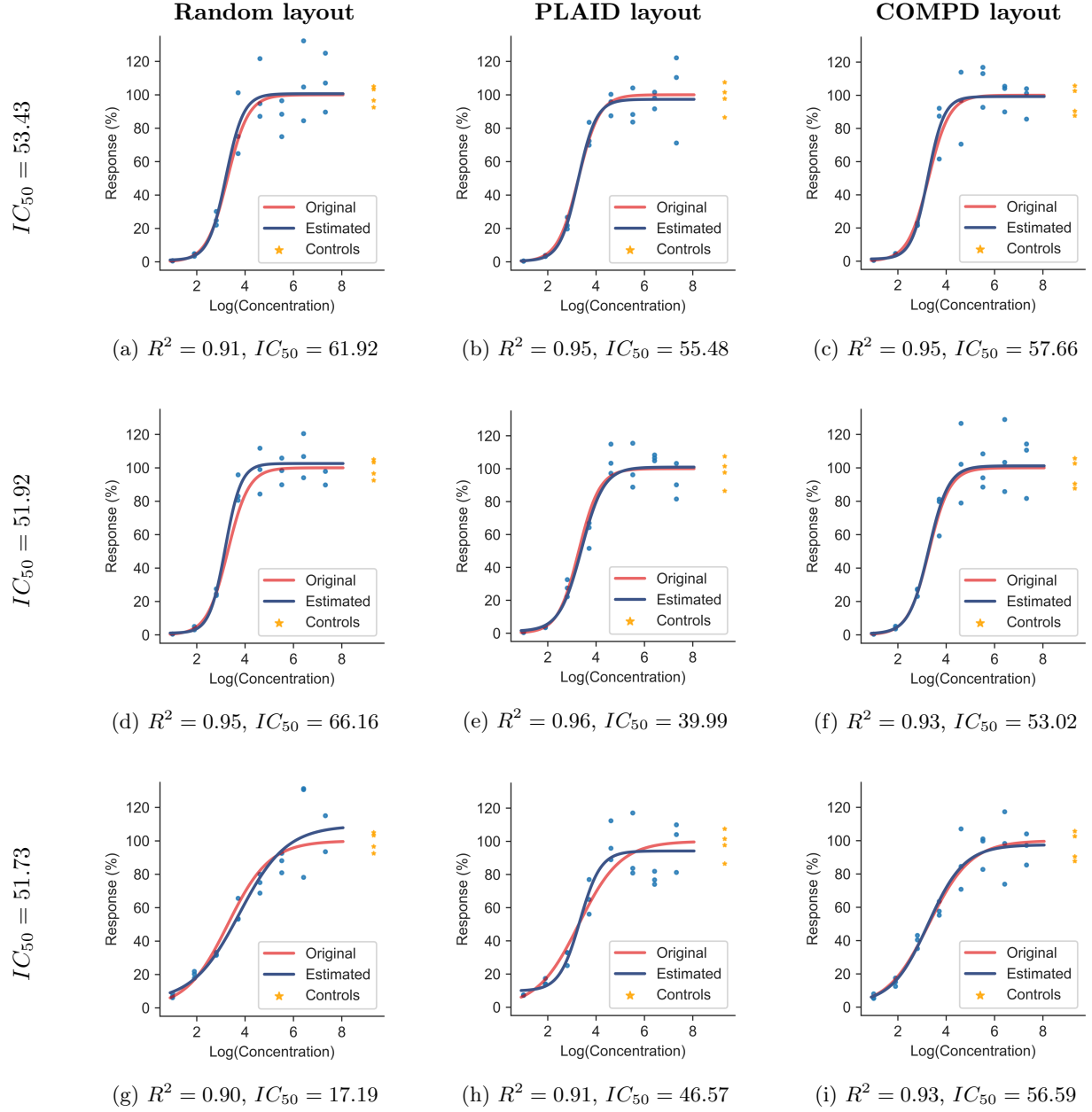


Figure 3: Examples of dose response curves for simulated experiments using 8 doses with 3 replicates (blue dots), 4 negative controls (yellow stars), and strong bowl-shaped plate effects. On the left of each row we show the expected IC_{50} value. On the caption below each plot we show the obtained R^2 and IC_{50} values.

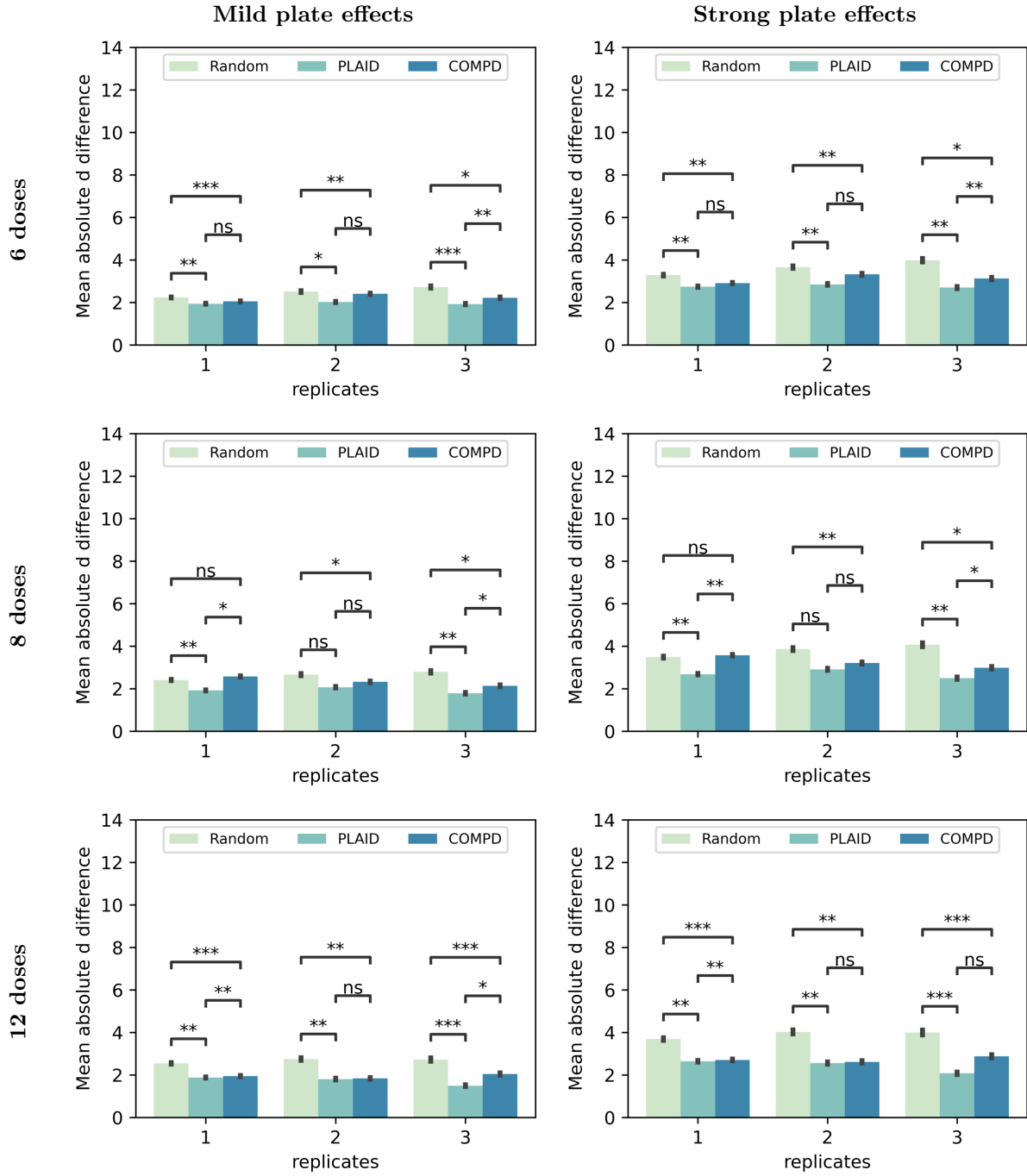


Figure 4: Mean absolute difference between expected and obtained heights (max value) for dose-response curves using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoidal curves were fitted using only the compound data (not including negative controls). *** indicates $p \leq 10^{-43}$.

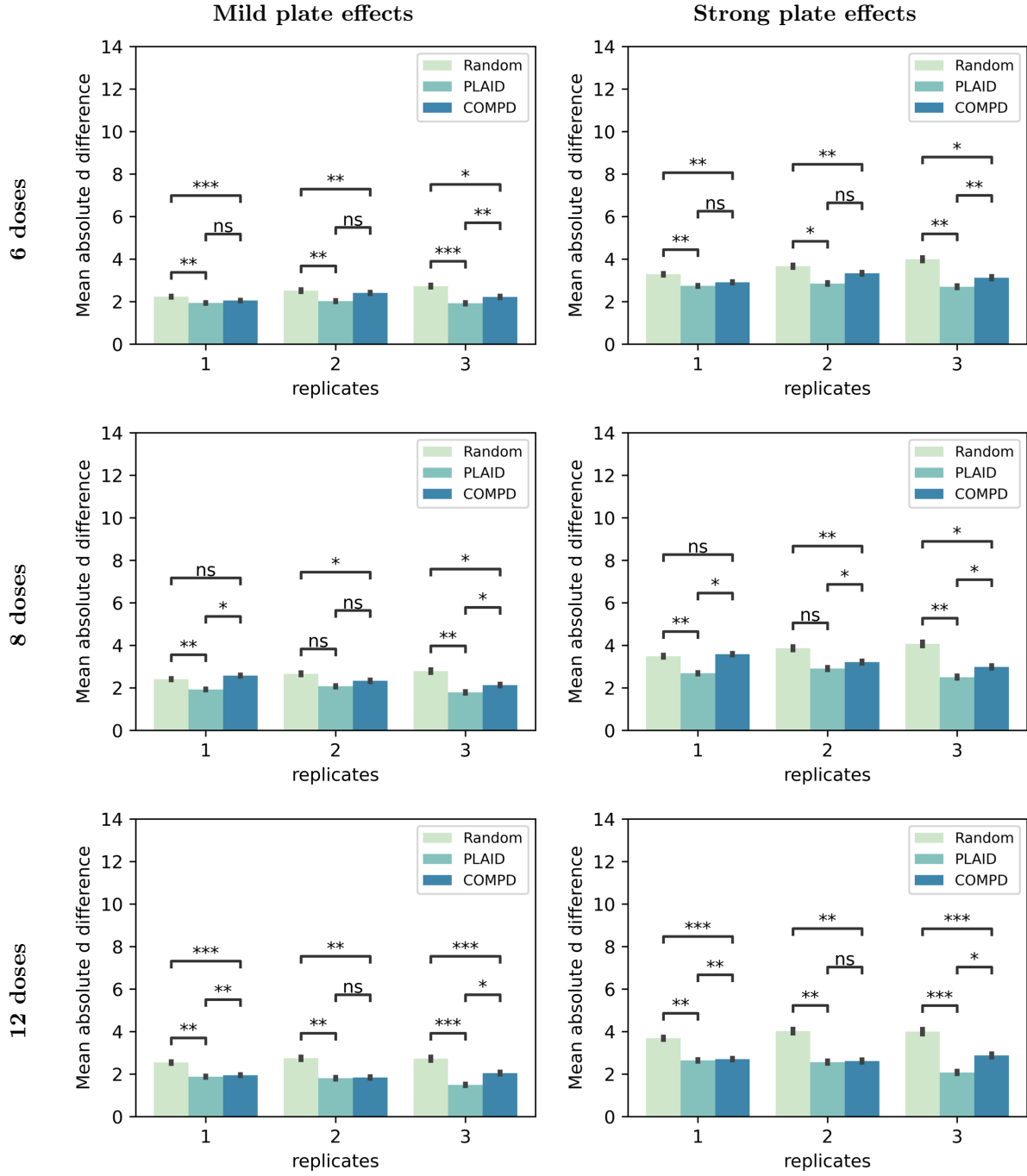


Figure 5: Mean absolute difference between expected and obtained heights (max value) for dose-response curves using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoidal curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate. *** indicates $p \leq 10^{-43}$.

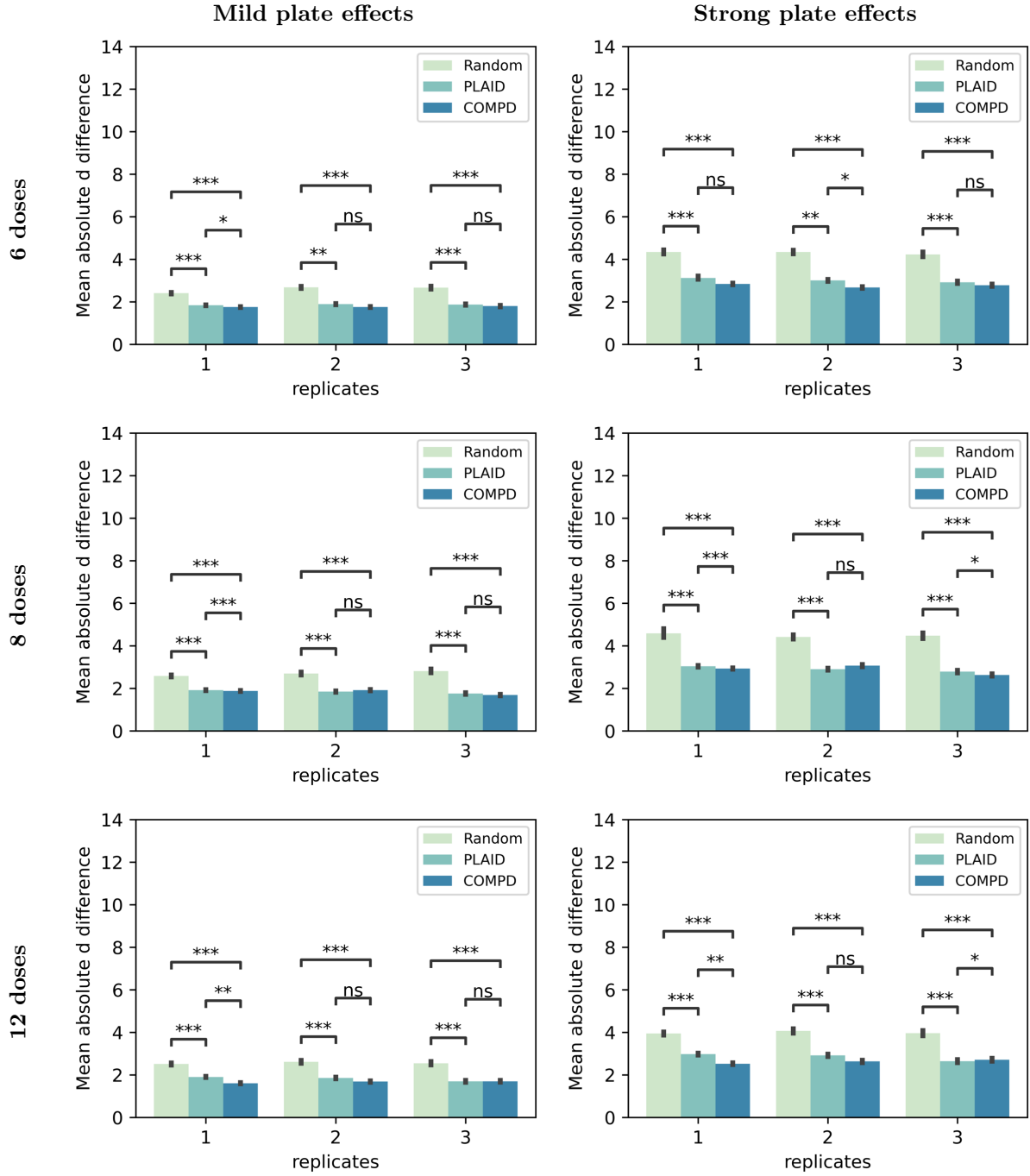


Figure 6: Mean absolute difference between expected and obtained heights (max value) for dose-response curves using various numbers of doses, replicates, and plate-effect strengths with a linear relationship to column number in the right half of the plate. The 4PL sigmoidal curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate. *** indicates $p \leq 10^{-43}$.

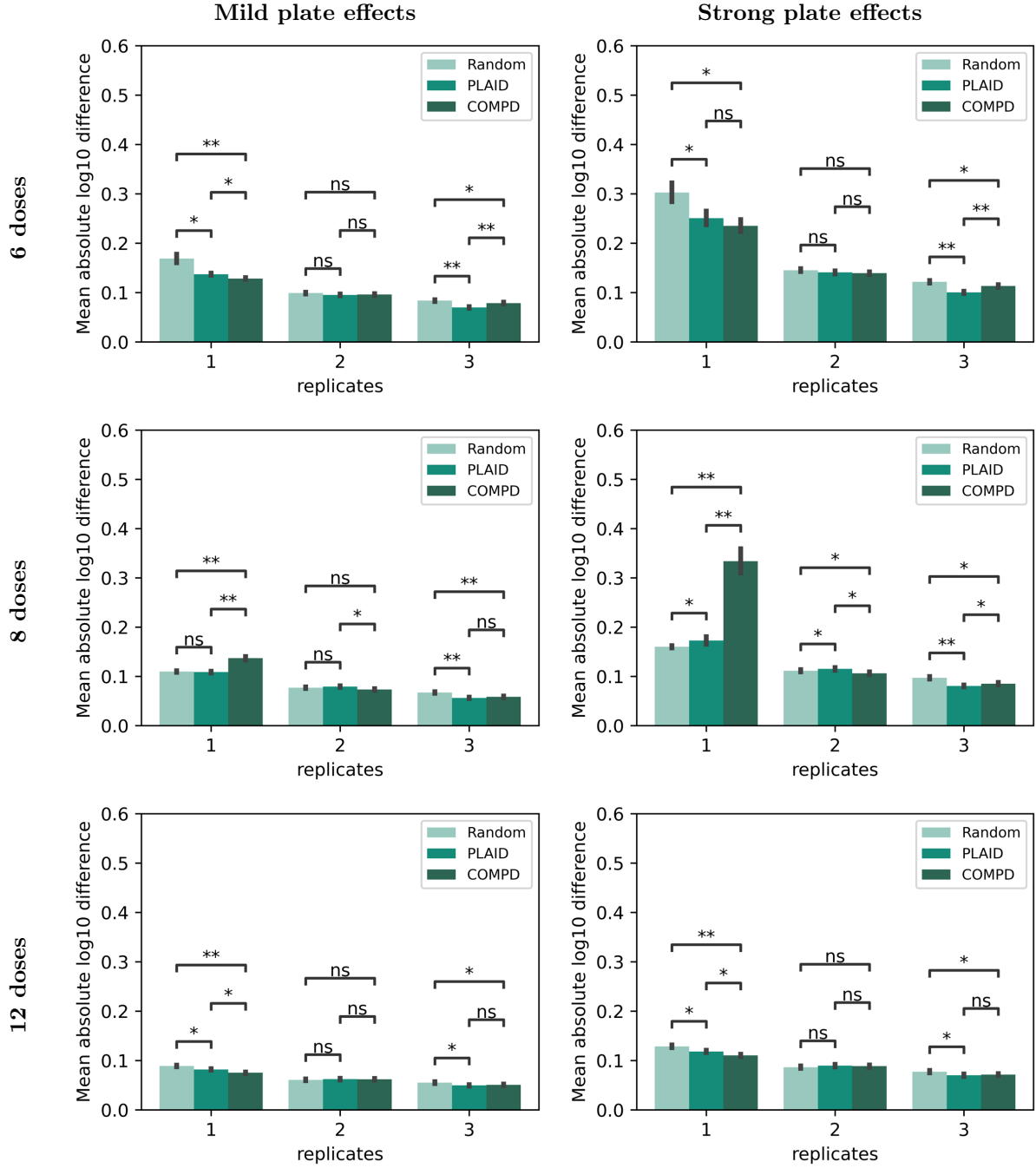


Figure 7: Mean absolute $\log_{10}$ difference between expected and obtained values of the relative EC_{50}/IC_{50} for dose-response curves using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoide curves were fitted using only compound data. * indicates $p \leq 0.05$, ** indicates $p \leq 10^{-12}$, *** indicates $p \leq 10^{-43}$.

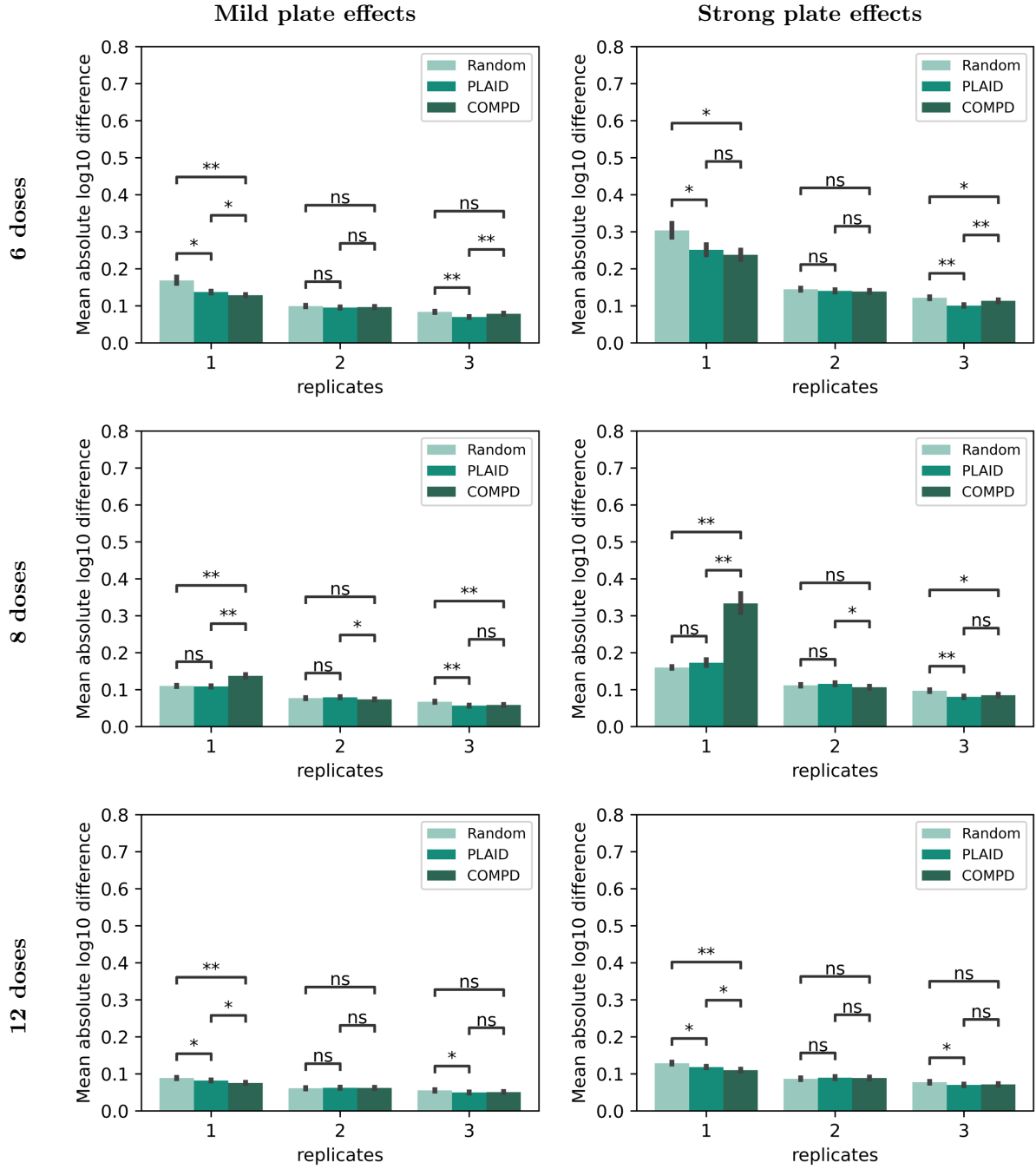


Figure 8: Mean absolute $\log_{10}$ difference between expected and obtained values of the relative EC_{50}/IC_{50} for dose-response curves using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoide curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate. * indicates $p \leq 10^{-4}$, ** indicates $p \leq 10^{-12}$, *** indicates $p \leq 10^{-43}$.

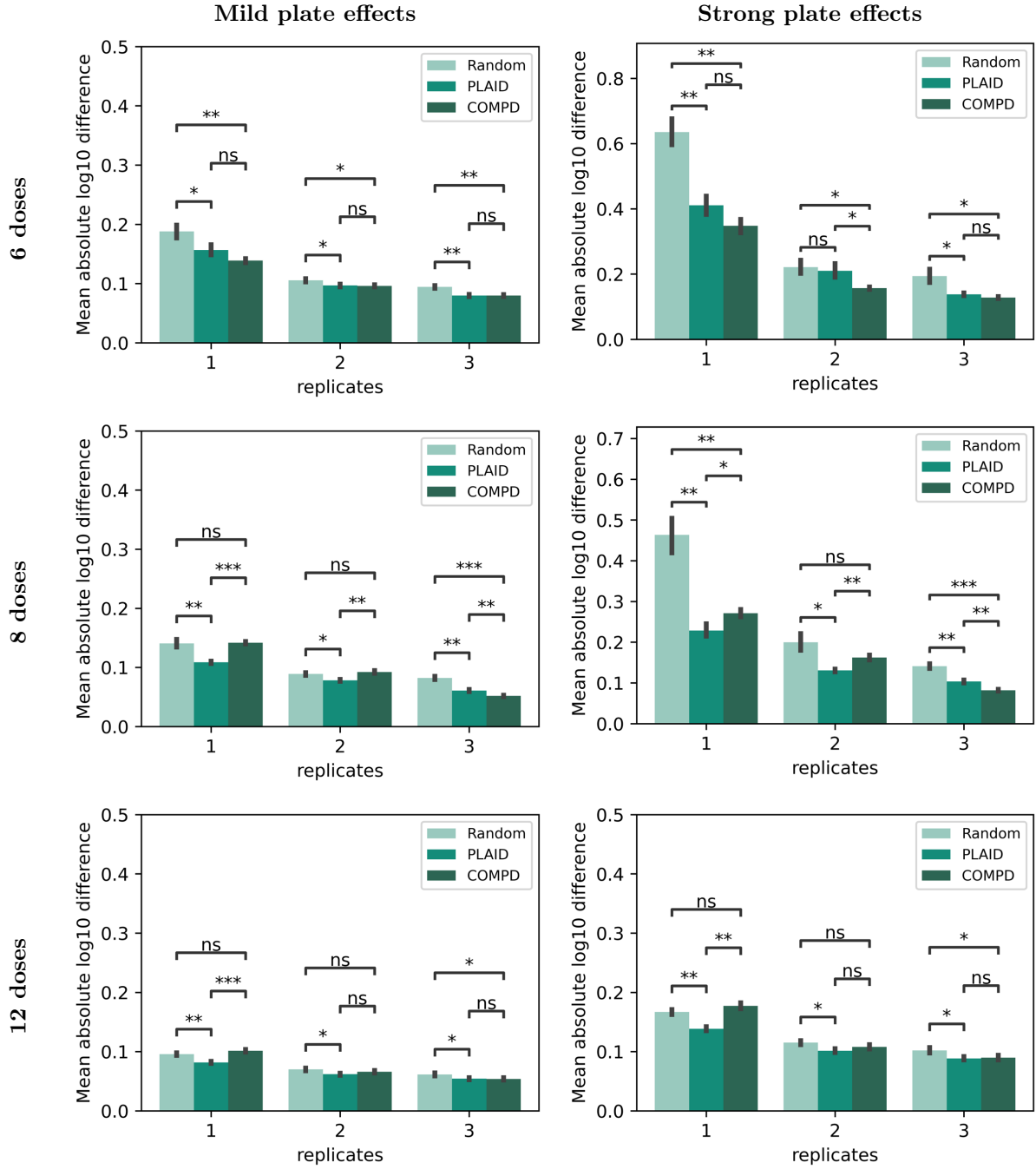


Figure 9: Mean absolute $\log_{10}$ difference between expected and obtained values of the relative EC_{50}/IC_{50} for dose-response curves using various numbers of doses, replicates, and plate-effect strengths with a linear relationship to column number in the right half of the plate. The 4PL sigmoide curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate. * indicates $p \leq 10^{-4}$, ** indicates $p \leq 10^{-12}$, *** indicates $p \leq 10^{-43}$.

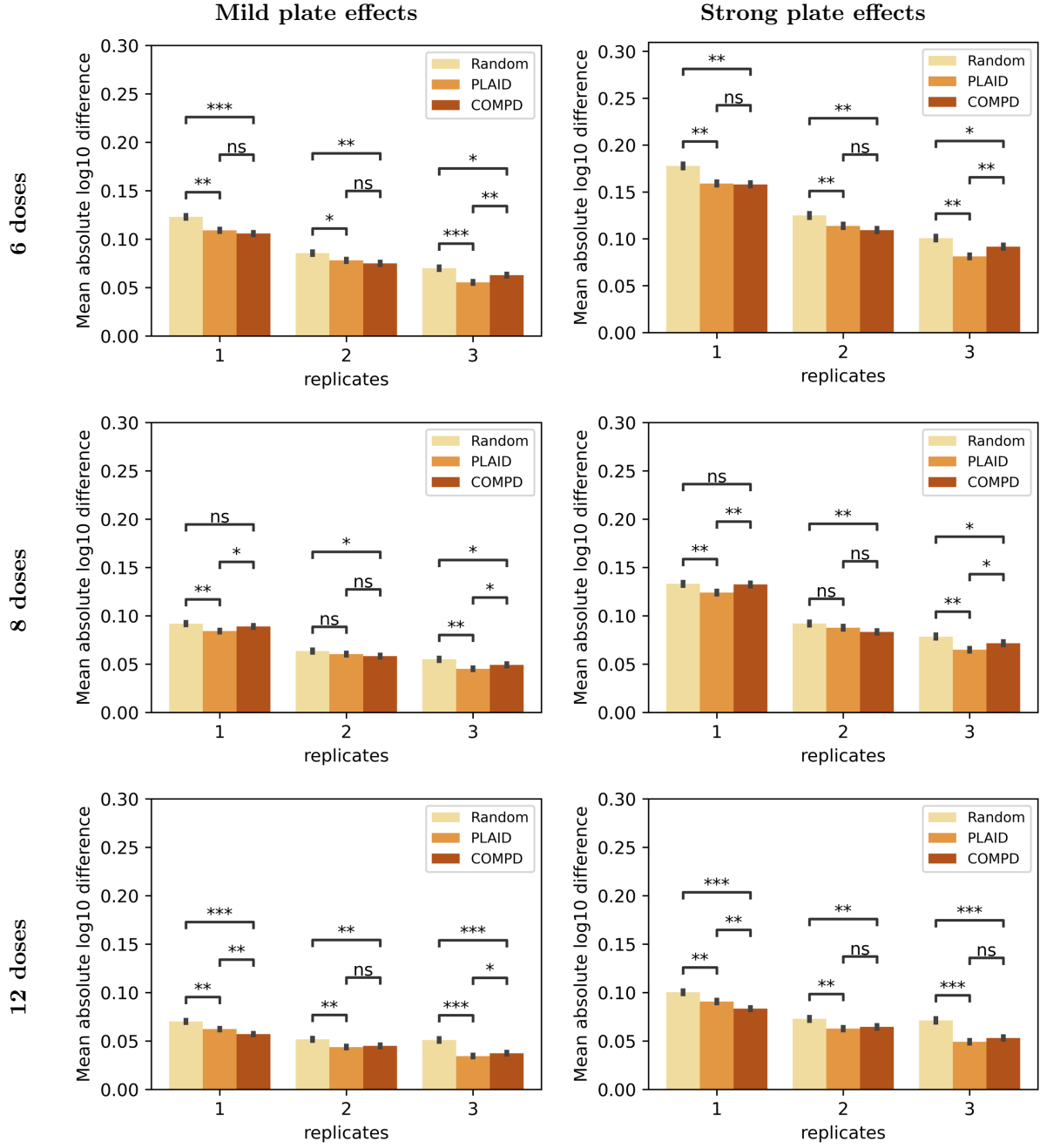


Figure 10: Mean absolute $\log_{10}$ difference between expected and obtained values of the absolute EC_{50}/IC_{50} for dose-response curves using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoidal curves were fitted using only compound data. *** indicates $p < 10^{-43}$.

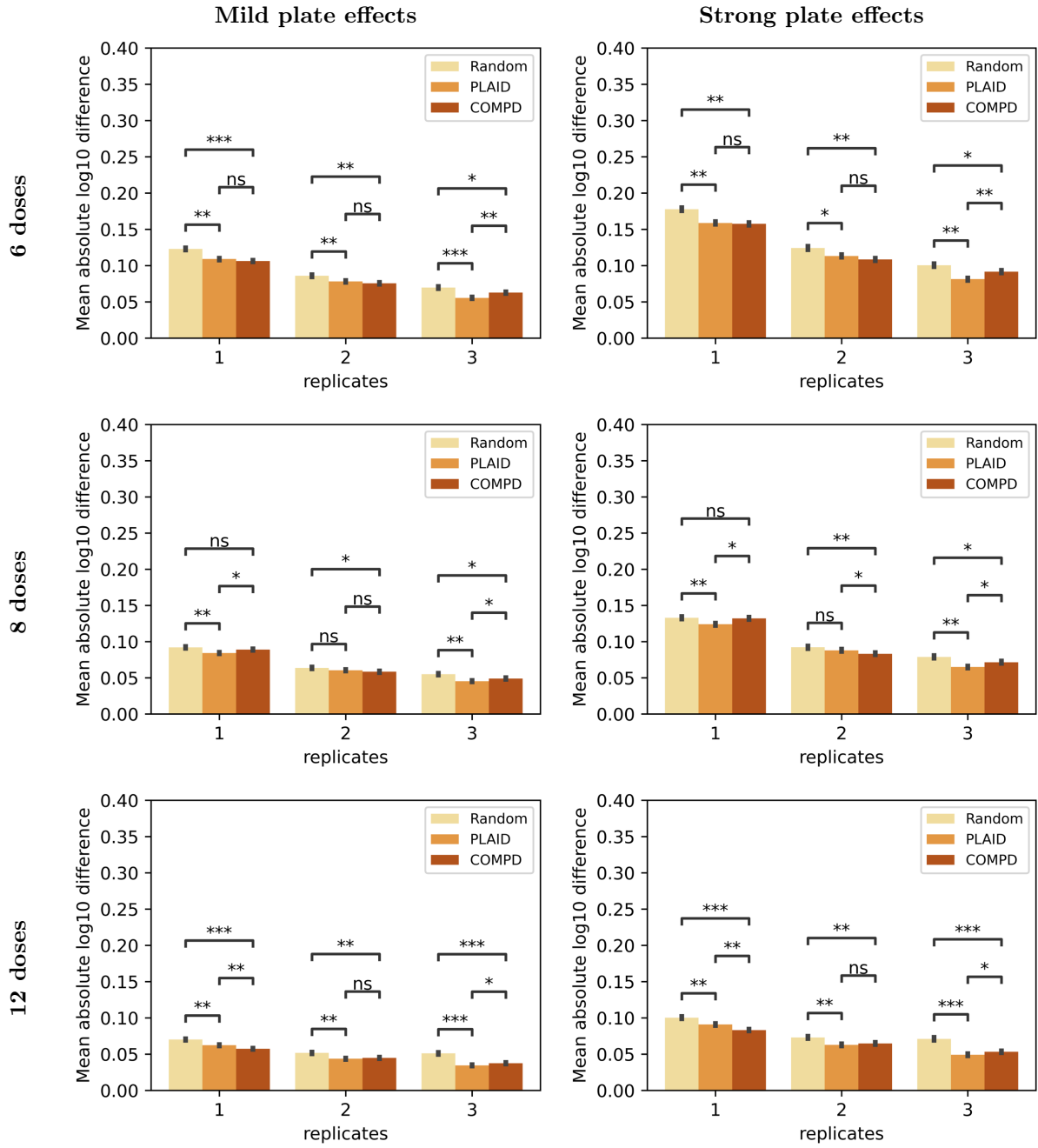


Figure 11: Mean absolute $\log_{10}$ difference between expected and obtained values of the absolute EC_{50}/IC_{50} for dose-response curves using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoide curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate. *** indicates $p \leq 10^{-43}$.

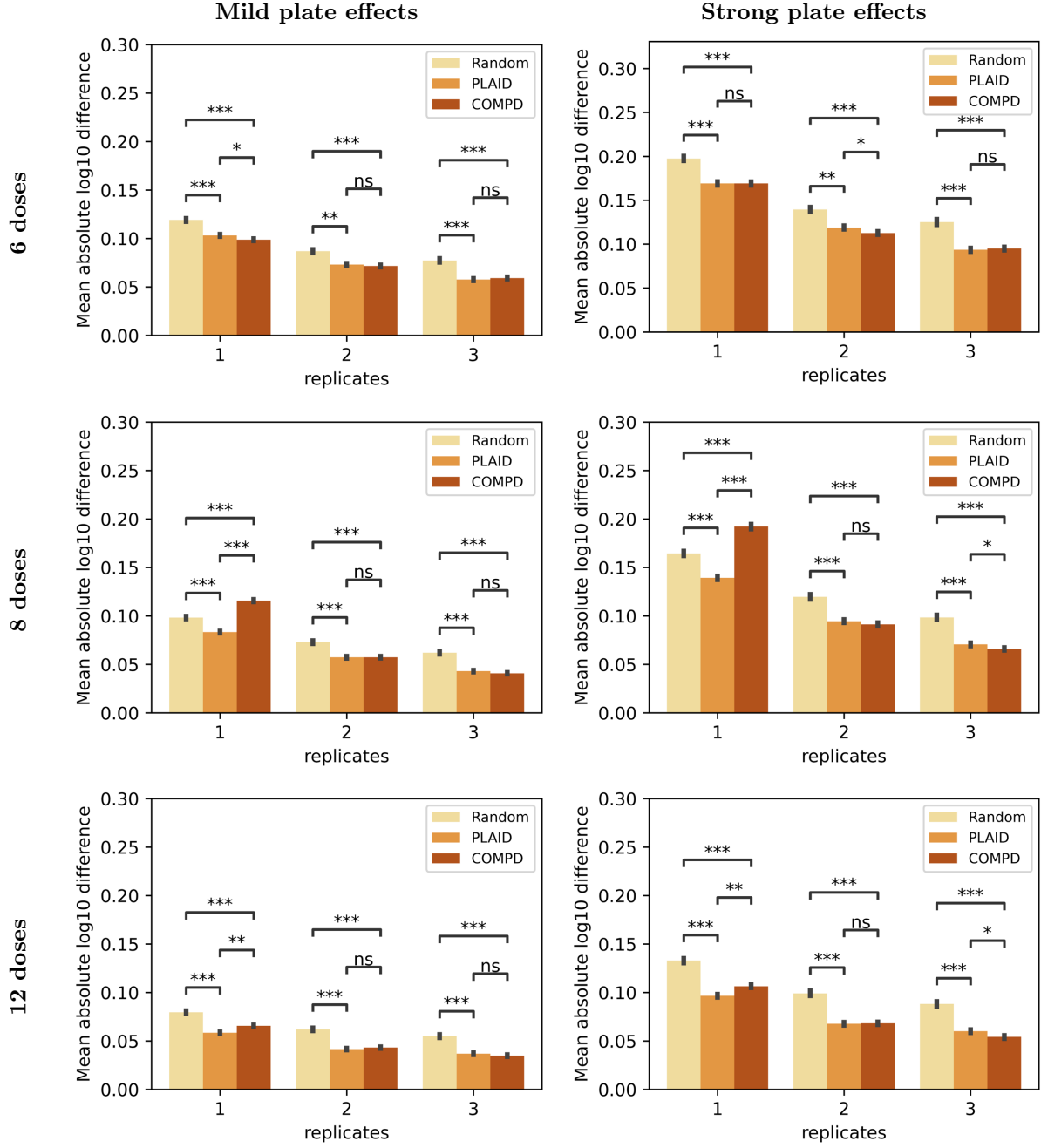


Figure 12: Mean absolute $\log_{10}$ difference between expected and obtained values of the absolute EC_{50}/IC_{50} for dose-response curves using various numbers of doses, replicates, and plate-effect strengths with a linear relationship to column number in the right half of the plate. The 4PL sigmoide curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate. *** indicates $p \leq 10^{-43}$.

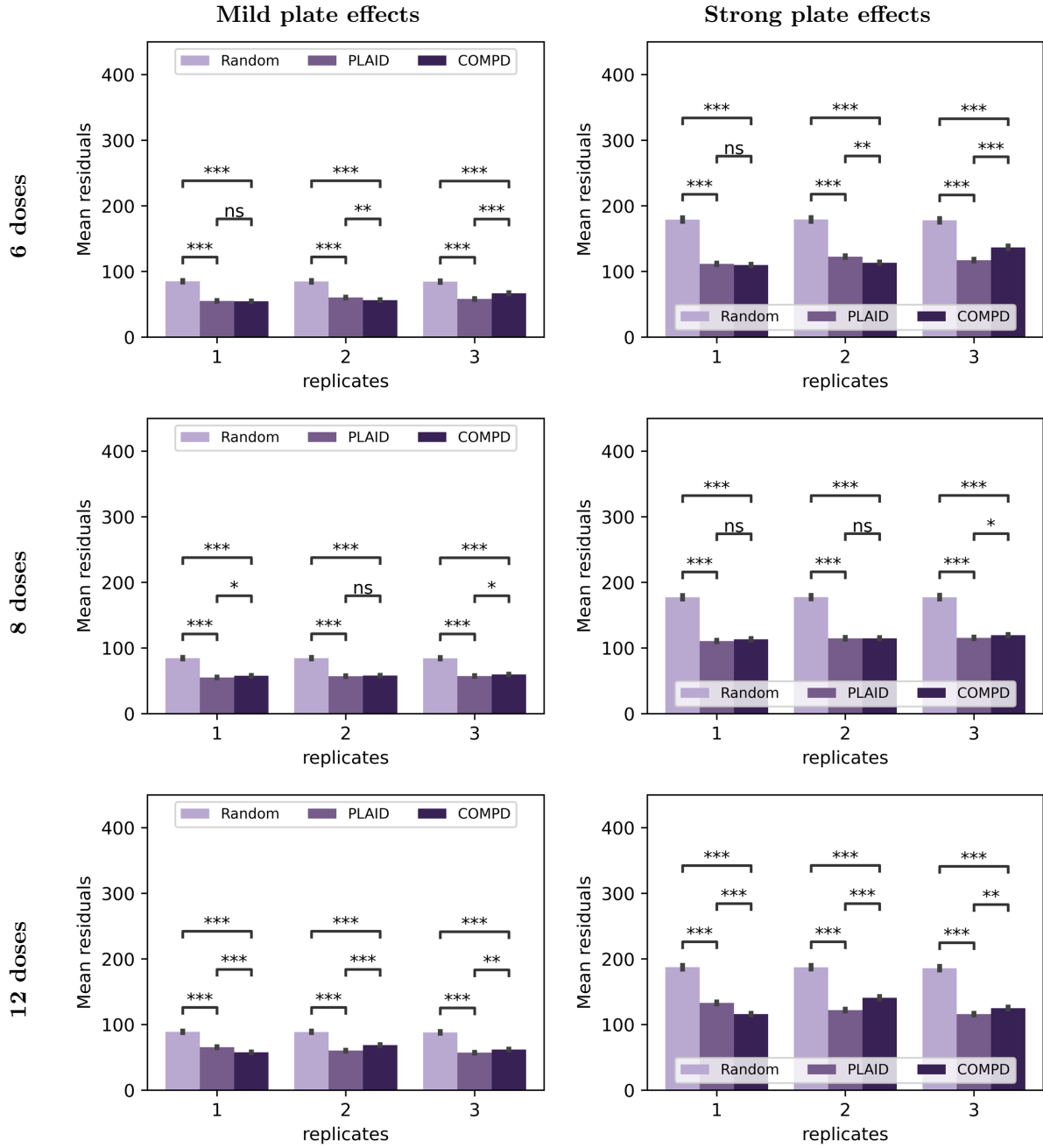


Figure 13: Residuals (MSE between expected and obtained responses) for dose-response simulations using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. *** indicates $p \leq 10^{-43}$.

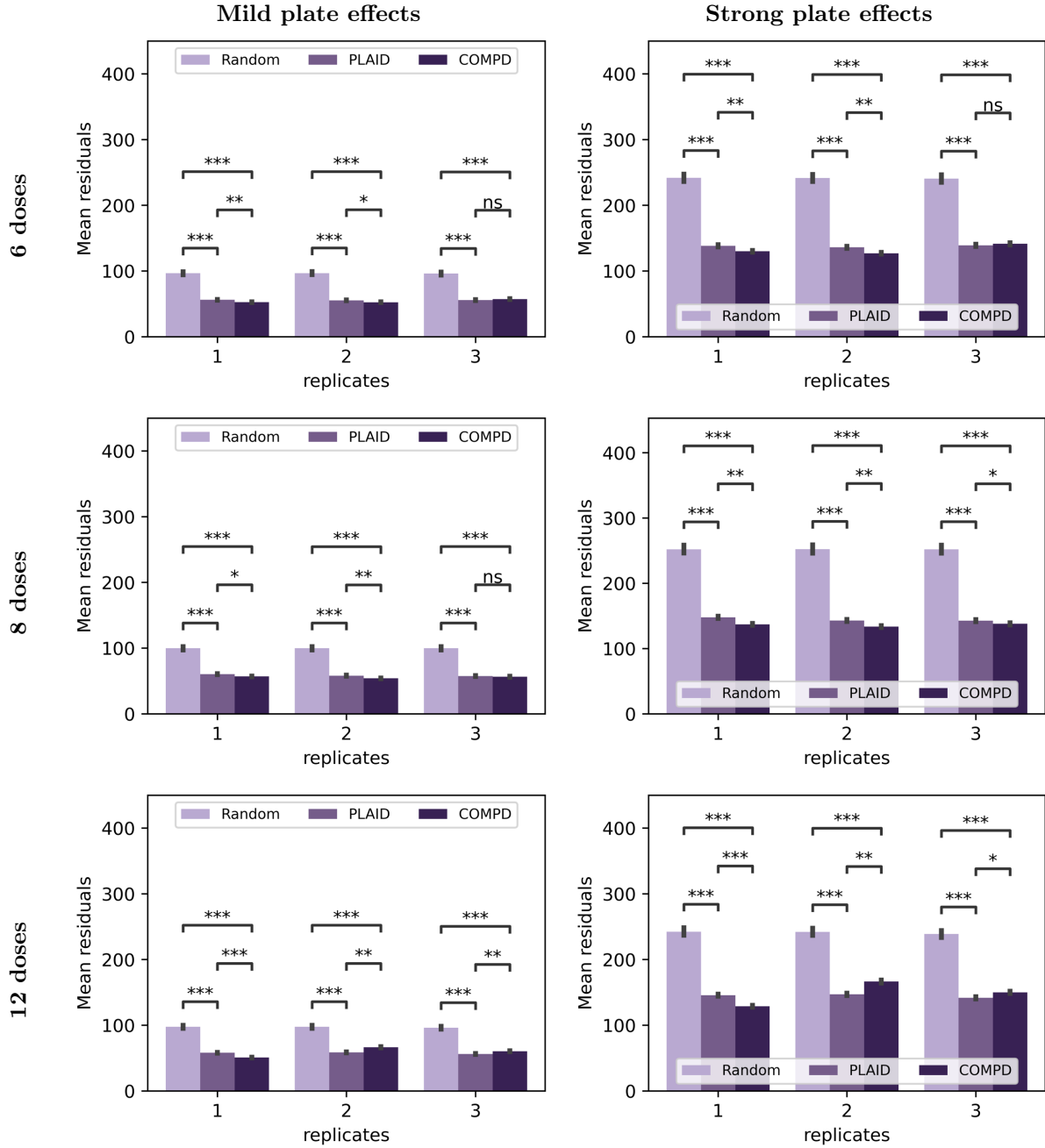


Figure 14: Residuals (MSE between expected and obtained responses) for dose-response simulations using various numbers of doses, replicates, and plate-effect strengths with a linear relationship to column number in the right half of the plate. *** indicates $p \leq 10^{-43}$.

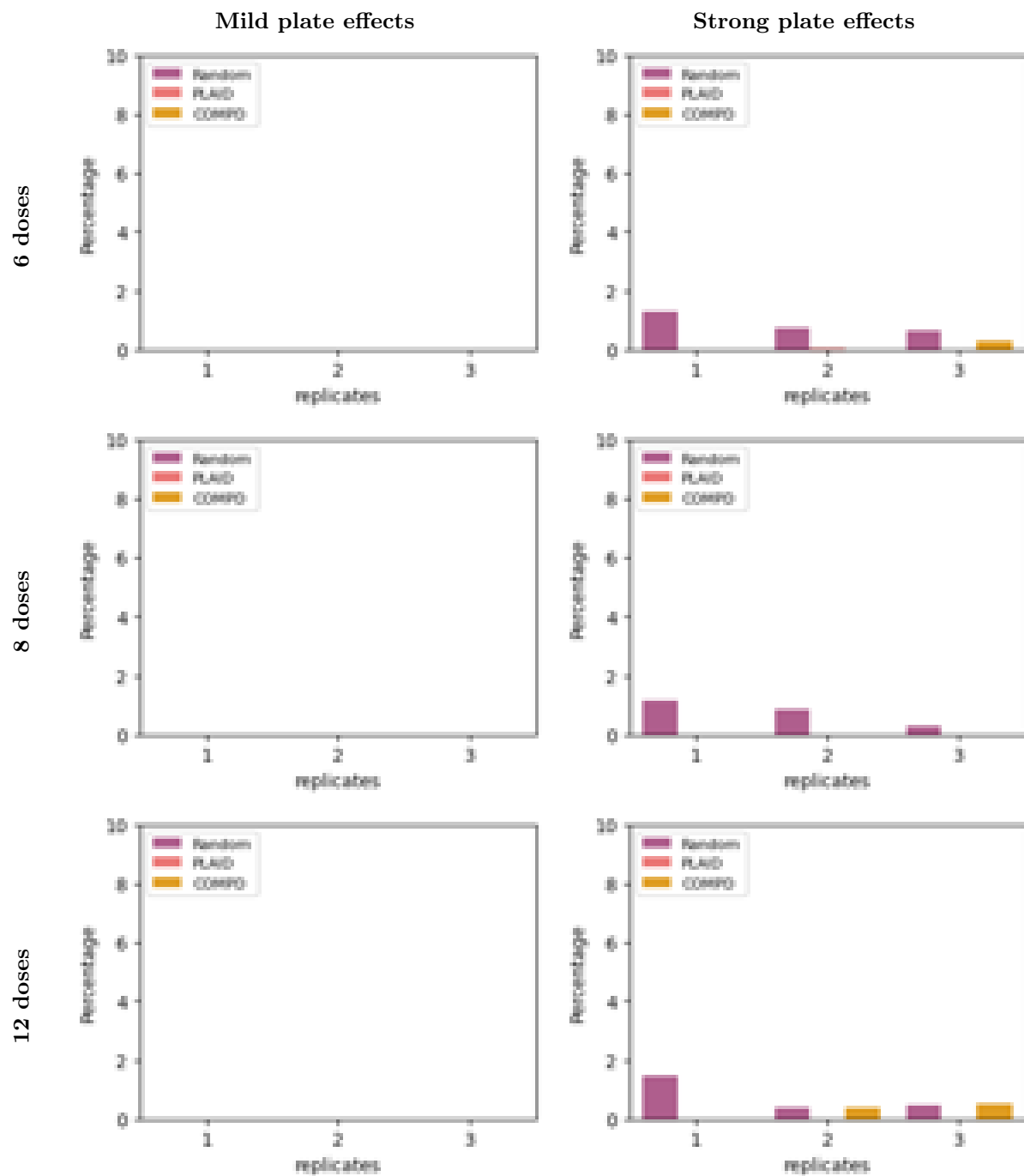


Figure 15: Percentage of dose-response curves with low-quality ($R^2 < 80\%$) using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoidal curves were fitted using only compound data.

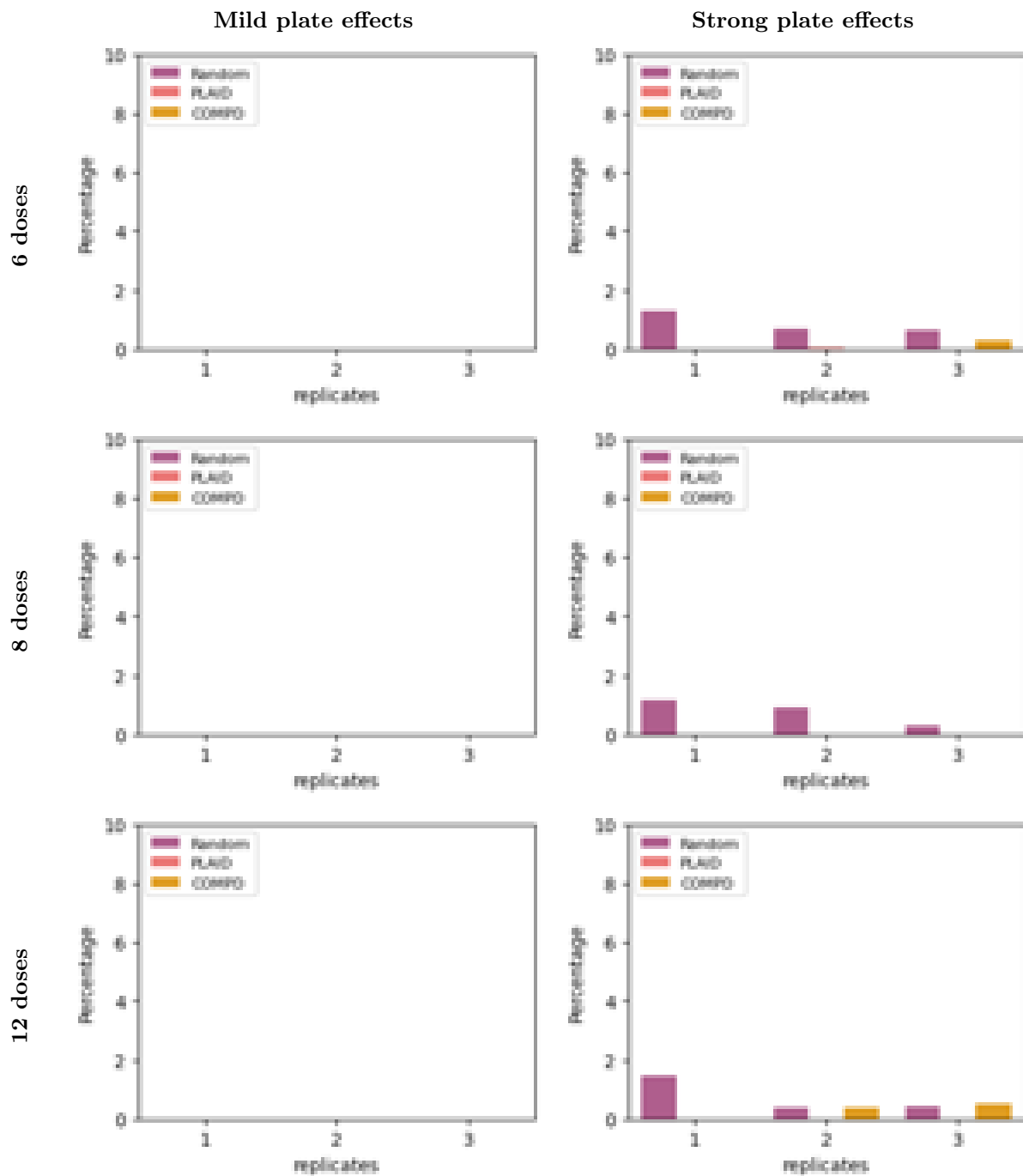


Figure 16: Percentage of dose-response curves with low-quality ($R^2 < 80\%$) using various numbers of doses, replicates, and bowl-shaped plate-effect strengths. The 4PL sigmoide curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate.

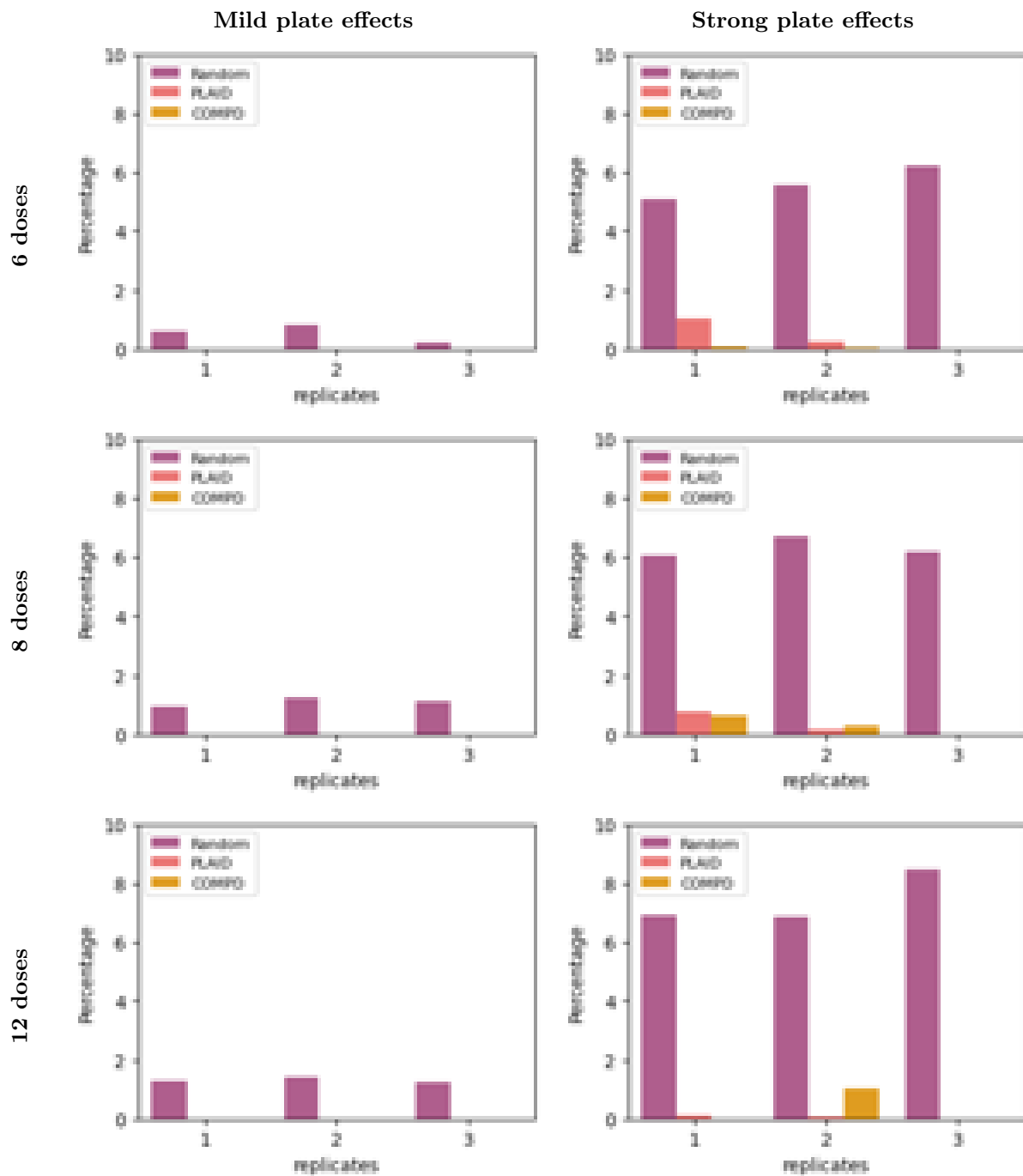


Figure 17: Percentage of dose-response curves with low-quality ($R^2 < 80\%$) using various numbers of doses, replicates, and plate-effect strengths with a linear relationship to column number in the right half of the plate. The 4PL sigmoide curves were fitted using compound data and 4 negative controls such that their mean was equal to the mean of the 20 negative controls the plate.

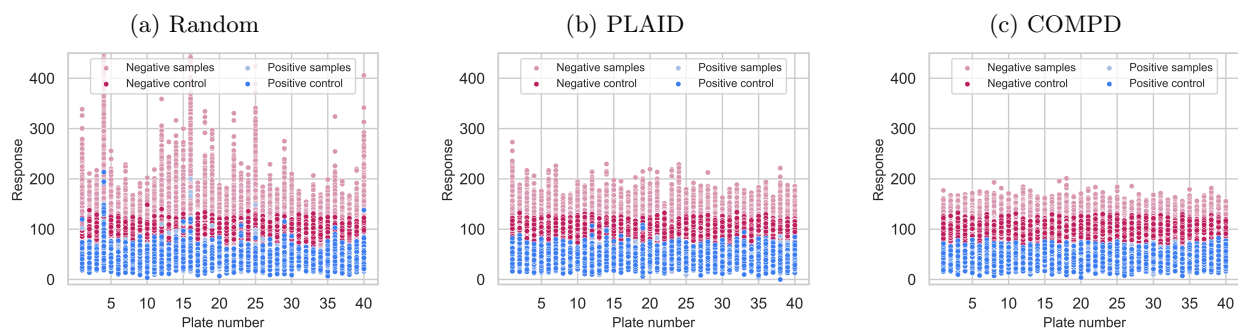


Figure 18: Comparison between expected and obtained values for screening experiments using 10 positive and 10 negative controls on 384-well plates with only 1 replicate, 1% hit-rate and mild bowl-shaped plate effects.

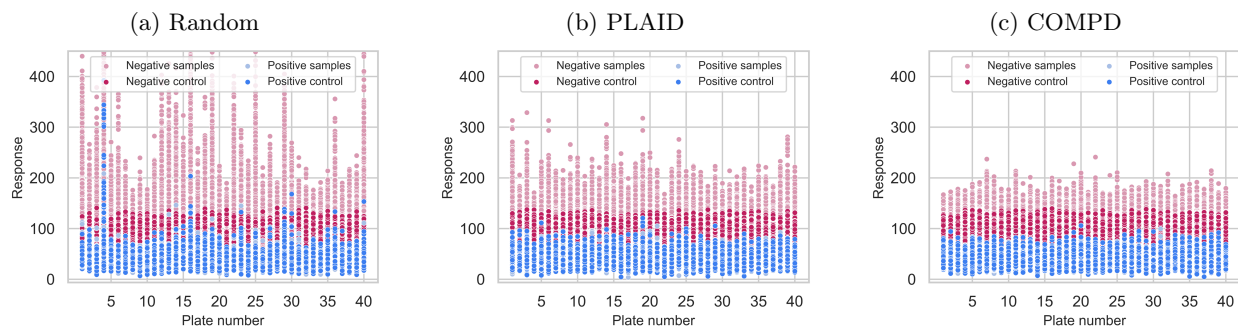


Figure 19: Comparison between expected and obtained values for screening experiments using 10 positive and 10 negative controls on 384-well plates with only 1 replicate, 1% hit-rate and moderate bowl-shaped plate effects.

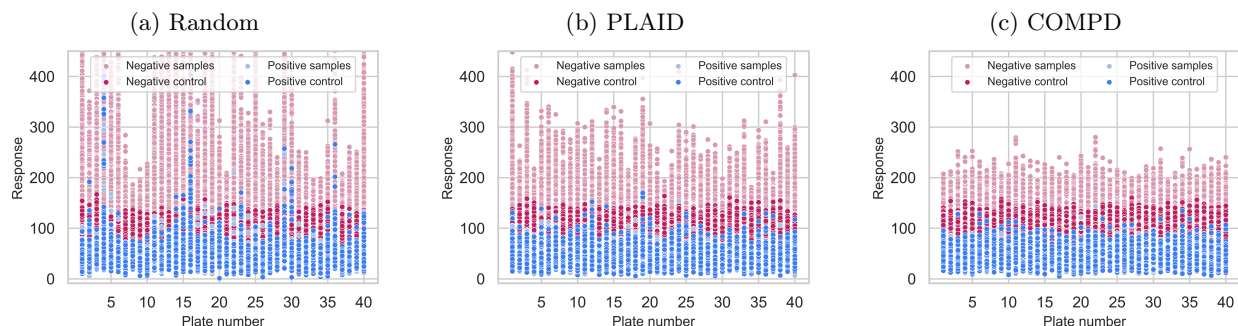
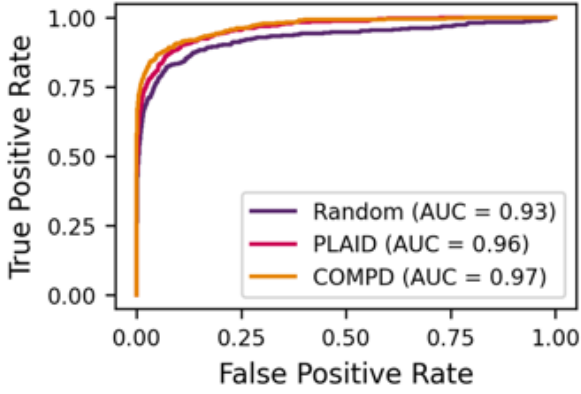
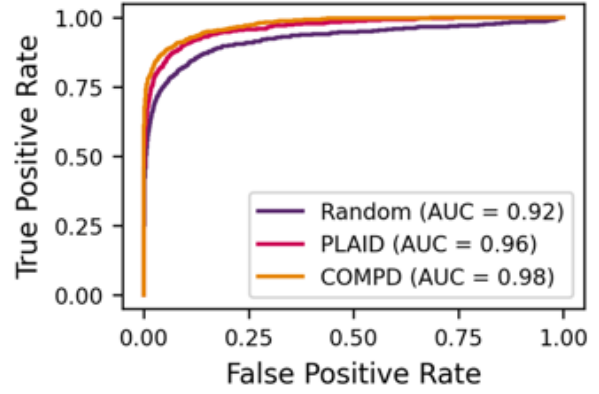


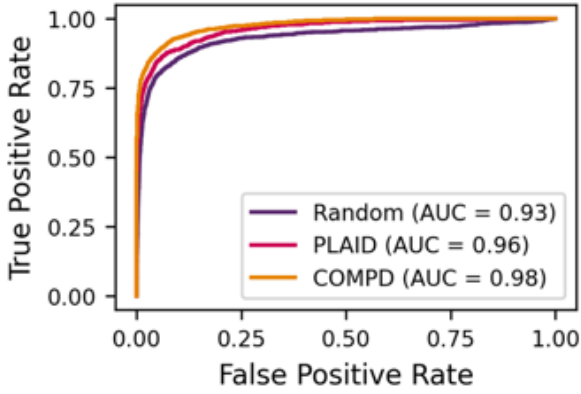
Figure 20: Comparison between expected and obtained values for screening experiments using 10 positive and 10 negative controls on 384-well plates with only 1 replicate, 1% hit-rate and strong bowl-shaped plate effects.



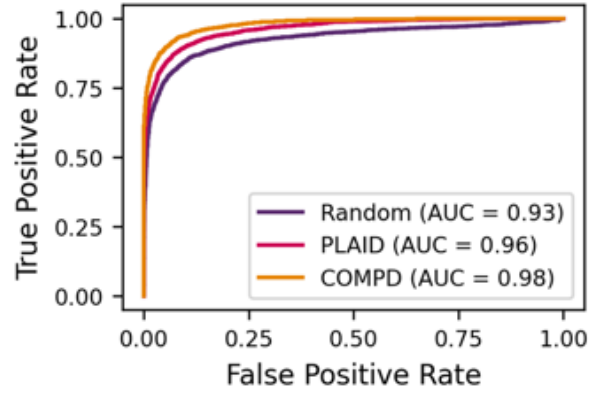
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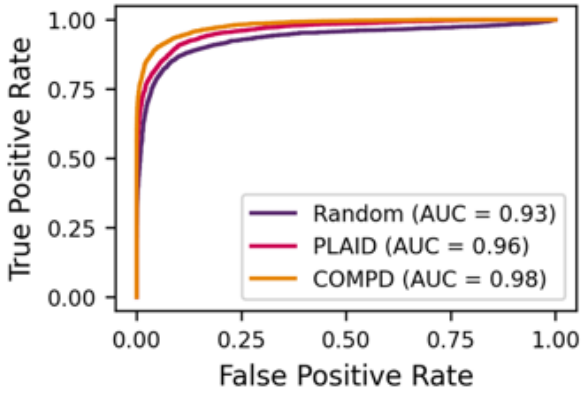
(b) 5% hits



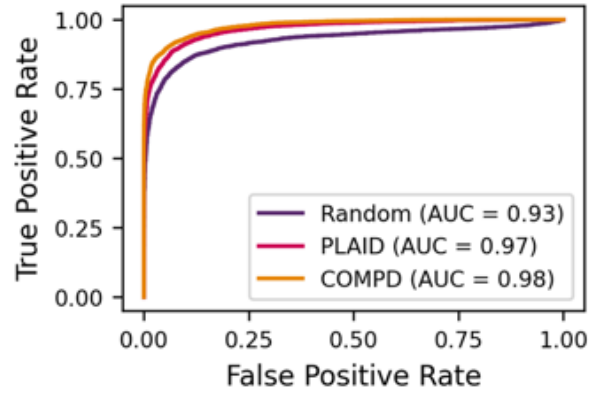
(c) 10% hits



(d) 20% hits

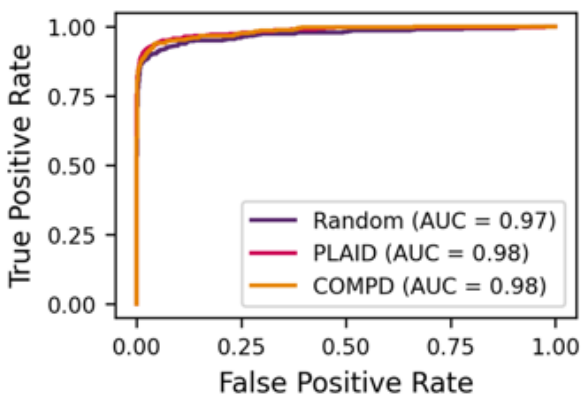


(e) 30% hits

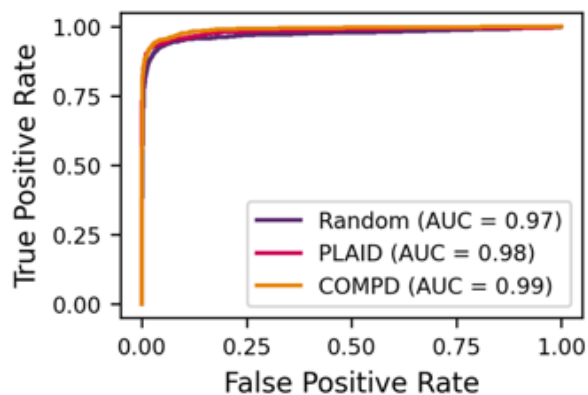


(f) 40% hits

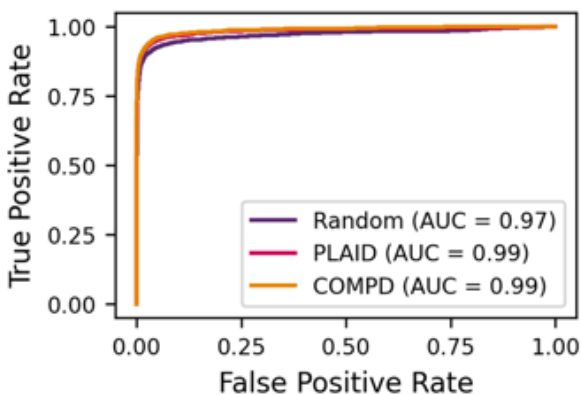
Figure 21: Comparison of ROC curves for screening experiments with different hit rates, using layouts with 10 positive and 10 negative controls in the presence of strong bowl-shaped plate effects.



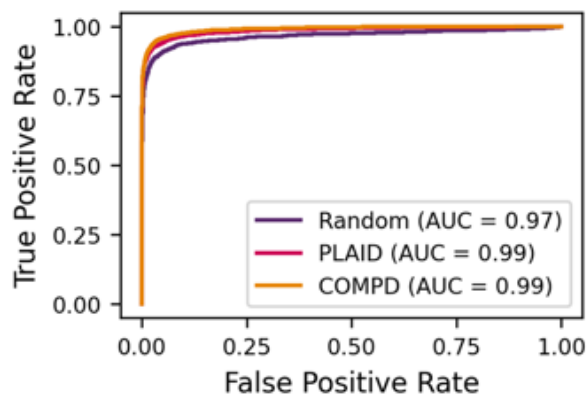
(a) 1% hits



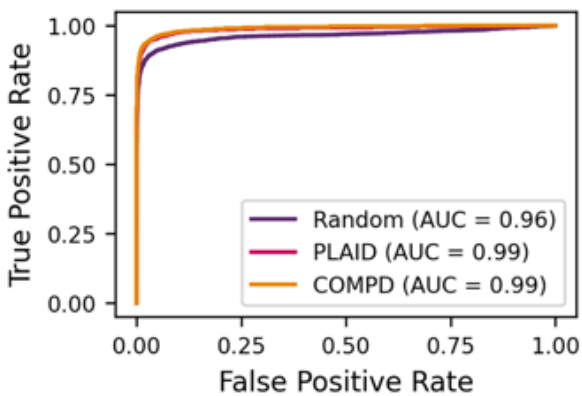
(b) 5% hits



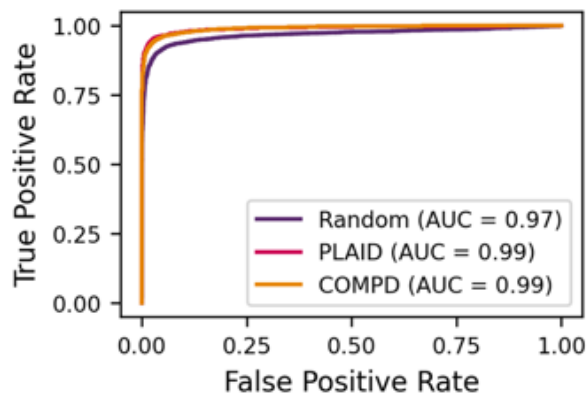
(c) 10% hits



(d) 20% hits



(e) 30% hits



(f) 40% hits

Figure 22: Comparison of ROC curves for screening experiments with different hit rates, using layouts with 8 positive and 8 negative controls in the presence of moderate bowl-shaped plate effects.

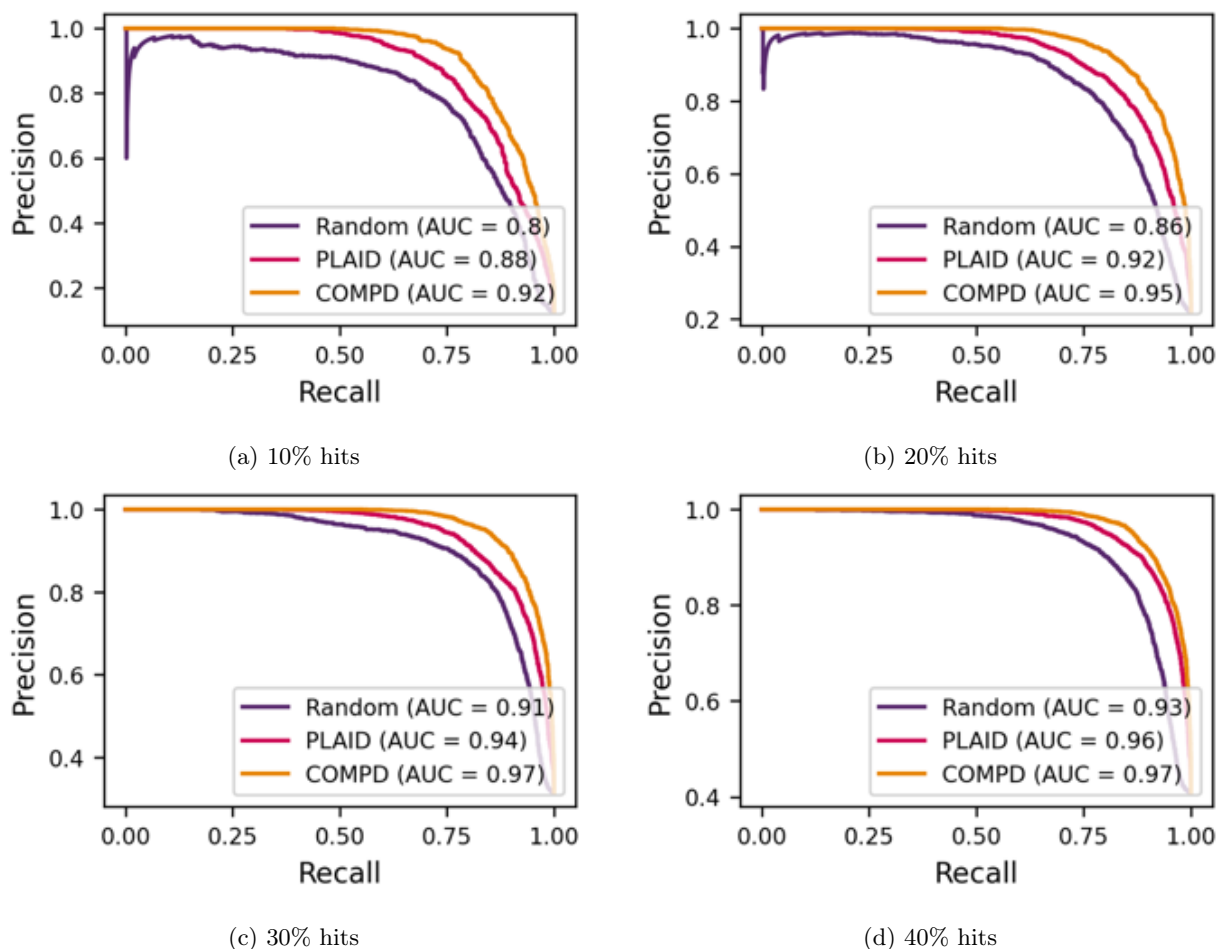
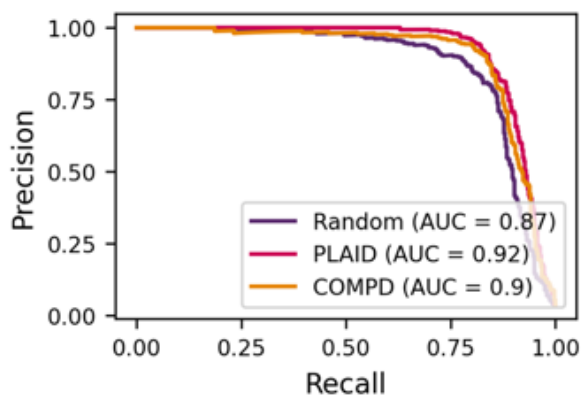


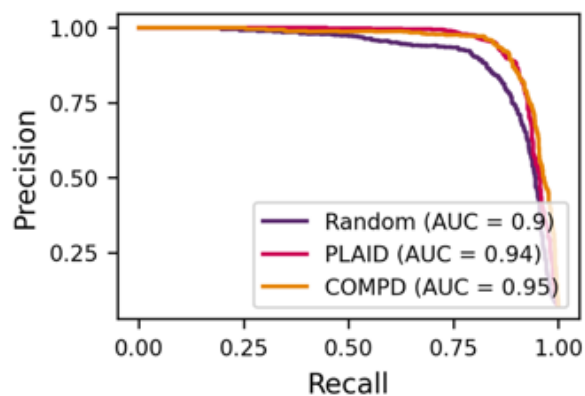
Figure 23: Comparison of PR curves for screening experiments with different hit rates, using layouts with 10 positive and 10 negative controls in the presence of strong bowl-shaped plate effects.

Measurement	Comparison	1 replicate	2 replicates	3 replicates
Relative IC_{50}/EC_{50}	Random – PLAID	5.18e-25	2.73e-10	6.72e-28
	Random – COMPD	1.26e-18	8.20e-04	2.98e-82
	PLAID – COMPD	3.20e-06	9.21e-20	2.39e-24
Absolute IC_{50}/EC_{50}	Random – PLAID	3.17e-57	1.27e-57	6.22e-72
	Random – COMPD	1.05e-53	1.50e-74	7.23e-103
	PLAID – COMPD	5.81e-231	1.04e-02	3.72e-05

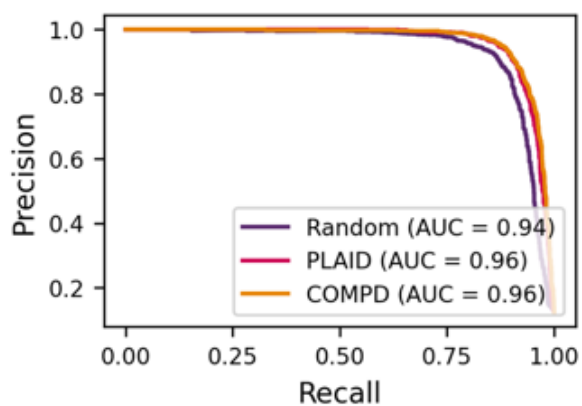
Table 1: p-values for the obtained relative IC_{50}/EC_{50} and absolute IC_{50}/EC_{50} for dose response curves with 8 doses, 1, 2, or 3 replicates, 20 negative controls on 384-well plate, and strong plate effects with a linear relationship to column number on the right side of the plate.



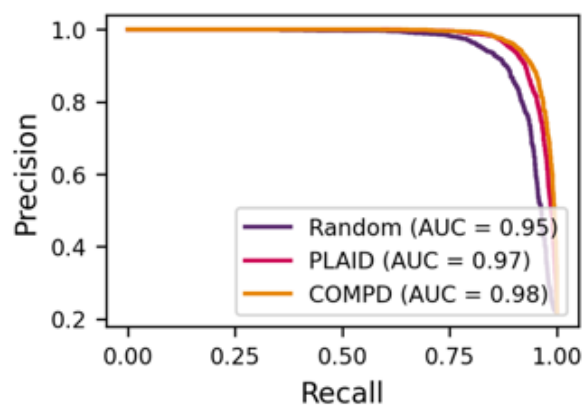
(a) 1% hits



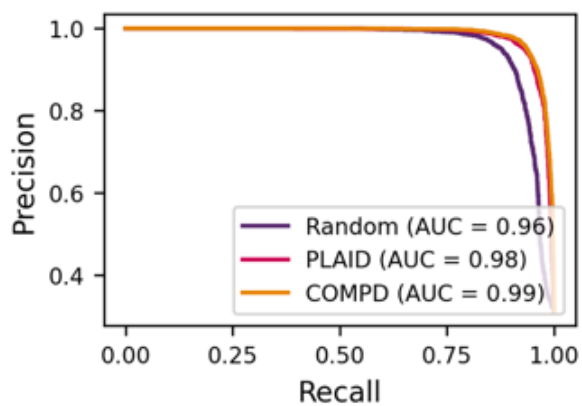
(b) 5% hits



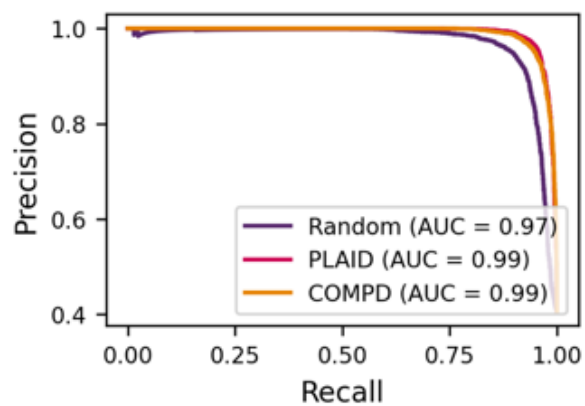
(c) 10% hits



(d) 20% hits



(e) 30% hits



(f) 40% hits

Figure 24: Comparison of PR curves for screening experiments with different hit rates, using layouts with 8 positive and 8 negative controls in the presence of moderate bowl-shaped plate effects.

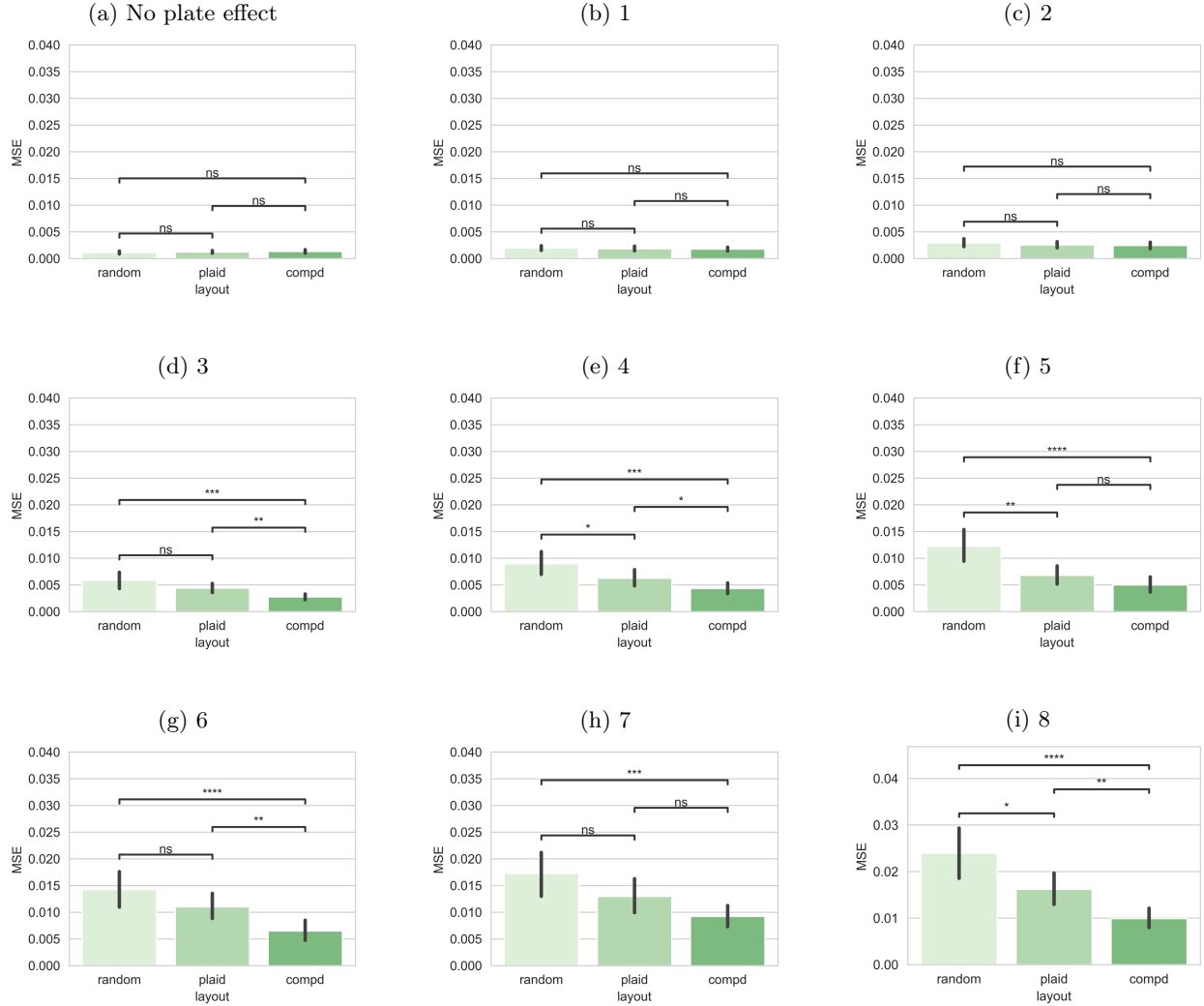


Figure 25: Comparison of MSE when calculating the SSMD of plates for screenings experiments using 10 positive and 10 negative controls on 384-well plate with only 1 replicate, 1% hit-rate and varying strengths of bowl-shaped plate effects. * indicates $p < 0.05$, ** indicates $p < 10^{-2}$, *** indicates $p < 10^{-3}$.

Hit-Rate	1%	5%	10%	20%	30%	40%
Random	0.93 $\pm$ (0.004)	0.93 $\pm$ (0.007)	0.93 $\pm$ (0.004)	0.93 $\pm$ (0.003)	0.93 $\pm$ (0.002)	0.93 $\pm$ (0.003)
PLAID	0.96 $\pm$ (0.002)	0.96 $\pm$ (0.004)	0.96 $\pm$ (0.002)	0.96 $\pm$ (0.002)	0.96 $\pm$ (0.002)	0.97 $\pm$ (0.002)
COMPd	0.97 $\pm$ (0.002)	0.98 $\pm$ (0.002)	0.98 $\pm$ (0.002)	0.98 $\pm$ (0.001)	0.98 $\pm$ (0.001)	0.98 $\pm$ (0.001)

Table 2: Mean and standard deviation of the ROC-AUC obtained for 10 screening experiments with varying hit rates, using layouts with 10 positive and 10 negative controls in the presence of very strong bowl-shaped plate effects. Each screening contains 40 plates. Hits are randomly located on the plates.

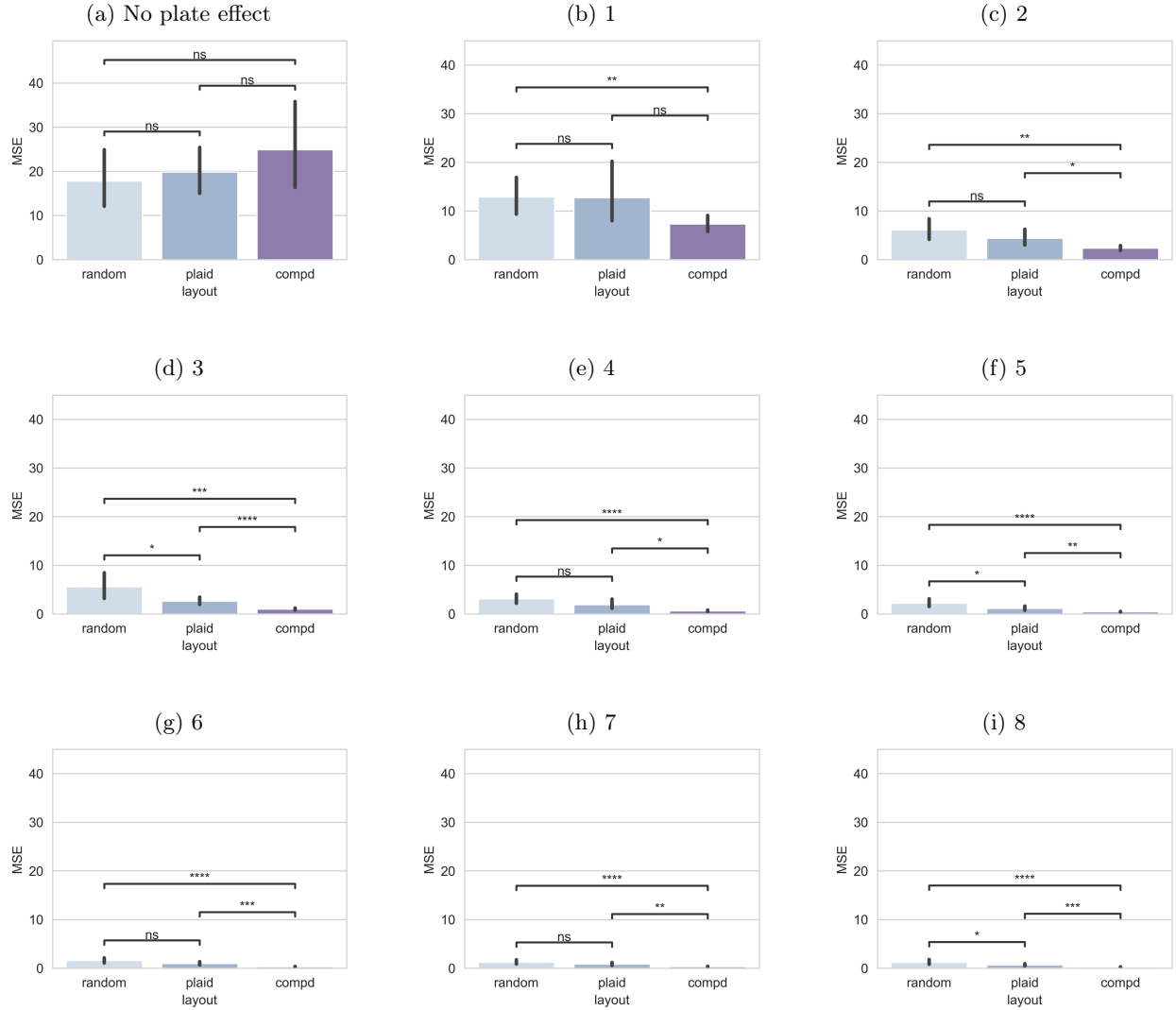


Figure 26: Comparison of MSE when calculating the SSMD of plates for screenings experiments using 10 positive and 10 negative controls on 384-well plate with only 1 replicate, 1% hit-rate and varying strengths of bowl-shaped plate effects. * indicates $p < 0.05$, ** indicates $p < 10^{-2}$, *** indicates $p < 10^{-3}$.

Hit rate	1%	5%	10%	20%	30%	40%
PLAID – Random	1.89×10^{-13}	1.37×10^{-09}	6.80×10^{-15}	3.48×10^{-16}	2.42×10^{-18}	9.75×10^{-18}
Random – COMPD	2.31×10^{-15}	3.51×10^{-10}	8.45×10^{-18}	5.85×10^{-13}	6.23×10^{-22}	1.02×10^{-20}
PLAID – COMPD	1.38×10^{-09}	4.98×10^{-08}	9.15×10^{-12}	4.92×10^{-10}	1.47×10^{-13}	2.66×10^{-14}

Table 3: p-values obtained when comparing the ROC-AUC obtained for 10 screening experiments with varying hit rates, using layouts with 10 positive and 10 negative controls in the presence of very strong bowl-shaped plate effects. Each screening contains 40 plates. Hits are randomly located on the plates.

Hit-Rate	1%	5%	10%	20%	30%	40%
Random	$0.68 \pm (0.032)$	$0.77 \pm (0.017)$	$0.82 \pm (0.017)$	$0.87 \pm (0.005)$	$0.91 \pm (0.003)$	$0.93 \pm (0.004)$
PLAID	$0.78 \pm (0.009)$	$0.84 \pm (0.008)$	$0.88 \pm (0.005)$	$0.92 \pm (0.004)$	$0.94 \pm (0.002)$	$0.96 \pm (0.002)$
COMPD	$0.84 \pm (0.007)$	$0.9 \pm (0.007)$	$0.92 \pm (0.005)$	$0.95 \pm (0.001)$	$0.97 \pm (0.002)$	$0.97 \pm (0.001)$

Table 4: Mean and standard deviation of the PR-AUC obtained for 10 screening experiments with varying hit rates, using layouts with 10 positive and 10 negative controls in the presence of very strong bowl-shaped plate effects. Each screening contains 40 plates. Hits are randomly located on the plates.

Hit rate	1%	5%	10%	20%	30%	40%
PLAID – Random	1.49×10^{-08}	2.11×10^{-10}	1.37×10^{-09}	1.48×10^{-15}	2.13×10^{-16}	1.76×10^{-15}
Random – COMPD	1.31×10^{-11}	2.31×10^{-14}	4.23×10^{-13}	2.31×10^{-20}	4.64×10^{-20}	8.23×10^{-19}
PLAID – COMPD	1.57×10^{-11}	5.73×10^{-12}	1.07×10^{-12}	4.89×10^{-15}	9.41×10^{-15}	6.75×10^{-14}

Table 5: p-values obtained when comparing the PR-AUC obtained for 10 screening experiments with varying hit rates, using layouts with 10 positive and 10 negative controls in the presence of very strong bowl-shaped plate effects. Each screening contains 40 plates. Hits are randomly located on the plates.